# Mock Exam 3 (Practice Only)

Name: _____ Section: ◯ 12 ◯ 29

Show all work. 17 questions, 6 points each (102 total). Scientific calculators only; no graphing calculators, phones, or sharing calculators.

**1.** Simplify: $\dfrac{x^3 + 7x^2 + 12x}{x^2 + 4x}$.

**2.** Evaluate: $64^{-\frac{1}{3}}$.

**3.** Rationalize the denominator and simplify: $\dfrac{4 - \sqrt{2}}{3 + \sqrt{3}}$.

**4.** Evaluate and express your answer in the form $a + bi$: $(8 + 5i) - (15 + 12i)$.

**5.** Let $f(x) = \dfrac{5x + 1}{3x - 6}$. Find the domain of $f$ in interval notation.

**6.** Simplify: $\dfrac{x-3}{x-5} - \dfrac{x-8}{x^2-25}$.

**7.** Simplify: $\dfrac{x^2+5x-24}{x^2-5x+6} \div \dfrac{x+8}{x+1}$.

**8.** Find the quotient and remainder using long division: $(3x^3 + 5x^2 - 4x + 7) \div (x + 2)$.

Quotient: _____ Remainder: _____

**9.** Simplify. Assume all variables are positive: $\sqrt[5]{9x^2} \cdot \sqrt[5]{3x^3}$.

**10.** Solve: $\sqrt{3x+1} - 2 = 3$.

**11.** Simplify the complex fraction: $\dfrac{\frac{3}{x-1} - 2x}{\frac{4}{x-1} + \frac{3}{x}}$.

**12.** Divide and simplify. Write your answer in the form $a + bi$: $\dfrac{3 + 5i}{2 - i}$.

**13.** Simplify completely: $6\sqrt{75} + 4\sqrt{48}$.

**14.** Avery can paint a room in 8 hours. Working together with Morgan, they can paint it in 5 hours. How long would Morgan take alone?

**15.** The number of hours needed to build a fence varies inversely with the number of people working. If 3 people take 48 hours, how long will 12 people take?

**16.** If a 150-pound person can have 900 mg of a medicine in a day, how much can a 110-pound person have?

**17.** Taylor can bicycle 45 miles in the same time as walking 15 miles. Taylor bicycles 10 mph faster than walking. How fast does Taylor walk?